

ALEXIS MARCHAND

📍 Instytut Matematyczny Polskiej Akademii Nauk
ul. Śniadeckich 8, 00-656 Warszawa, Poland

✉ amarchand@impan.pl

🌐 alexis-marchand.github.io

EMPLOYMENT

from 9/2025 to 9/2027	IM PAN, Warszawa (Poland) Adiunkt (Assistant Professor) Research project: <i>Analysis on groups</i> (NCN Maestro 13) Mentor: PIOTR NOWAK
from 5/2026 to 7/2026	Universität Regensburg / Humboldt-Stiftung (Germany) Humboldt Research Fellow
from 5/2027 to 7/2027	Research project: <i>Invariants of groups and non-positive curvature</i> Mentor: CLARA LÖH
from 9/2024 to 9/2025	Kyōto University / JSPS (Japan) JSPS Postdoctoral Fellow Research project: <i>Stable commutator length in finitely presented groups</i> Mentor: KŌJI FUJIWARA

EDUCATION

from 10/2021 to 10/2024	University of Cambridge (U.K.) • Ph.D. in Pure Mathematics Thesis: <i>Computations and lower bounds for scl and the relative Gromov seminorm</i> Defended on: 15/7/2024 Funding: EPSRC PhD studentship Supervisor: HENRY WILTON
from 9/2017 to 8/2021	École Normale Supérieure de Lyon (France) • Dipl. ENS Funding: ENS studentship awarded by national competition (rank: 16)
from 9/2020 to 7/2021	Université Grenoble Alpes (France) • M.Sc. in Mathematics Mention Très Bien Thesis: <i>Automorphismes extérieurs de produits libres : revêtements abéliens caractéristiques et représentations libres</i> Supervisor: FRANÇOIS DAHMANI
from 10/2019 to 6/2020	University of Cambridge (U.K.) • M.A.St. in Mathematics Pass with Merit Essay: <i>Rigidity theorems for hyperbolic groups</i> Supervisor: HENRY WILTON
from 9/2017 to 7/2019	ÉNS de Lyon / Université Lyon 1 (France) • B.Sc. in Mathematics First two years of ENS curriculum in mathematics (L3+M1)

PUBLICATIONS (PEER-REVIEWED)

- [1] Alexis Marchand, From letter-quasimorphisms to angle structures and spectral gaps for scl, **Israel J. Math.** **273** (2026), no. 2, 937-967. DOI:[10.1007/s11856-026-2898-3](https://doi.org/10.1007/s11856-026-2898-3) • arXiv:[2402.13856](https://arxiv.org/abs/2402.13856)
- [2] Alexis Marchand, Bavard duality for the relative Gromov seminorm, **Enseign. Math.** **72** (2026), no. 1-2, 1-46. DOI:[10.4171/lem/1087](https://doi.org/10.4171/lem/1087) • arXiv:[2305.02149](https://arxiv.org/abs/2305.02149)
- [3] Alexis Marchand, Isometric embeddings of surfaces for scl, **Groups Geom. Dyn.** **19** (2025), no. 4, 1445-1478. DOI:[10.4171/ggd/845](https://doi.org/10.4171/ggd/845) • arXiv:[2302.04133](https://arxiv.org/abs/2302.04133)
- [4] Rémi Coulon and Alexis Marchand, Growth and co-growth in negatively curved groups, in Dahmani, Erschler, Horbez, Wise (eds.), Groups acting on fractals, hyperbolicity and auto-similarity, **Panoramas et Synthèses** **63** (2025).
- [5] Alexis Marchand, Free representations of outer automorphism groups of free products via characteristic abelian coverings, **J. Group Theory** **26** (2023), no. 2, 399-420. DOI:[10.1515/jgth-2021-0154](https://doi.org/10.1515/jgth-2021-0154) • arXiv:[2107.11230](https://arxiv.org/abs/2107.11230)

EXPOSITORY ARTICLES

- [6] Alexis Marchand, Makers of patterns: from Escher to Coxeter, **Eureka** (2020).
- [7] Alexis Marchand, Géométrie des groupes et courbure négative, **Journal de Mathématiques des Élèves de l'ENS de Lyon** (2020).

GRANTS & FUNDING

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| from 5/2026
to 7/2027 | Humboldt Stiftung • Humboldt fellowship's subsidy towards research costs
Allocated amount: EUR 4,800
Topic: <i>Invariants of groups and non-positive curvature</i> |
| from 9/2024
to 9/2025 | JSPS • Fellow's Research Support Allowance
Allocated amount: JPY 840,000
Topic: <i>Stable commutator length in finitely presented groups</i> |
| 6/2025 | RIMS, Kyōto University • RIMS Workshop (type A)
Allocated amount: JPY 350,000
Topic: <i>Spectrum problems for scl via S-machines</i> |

AWARDS

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| 3/2023 | Smith-Knight & Rayleigh-Knight prize • University of Cambridge
Grade 3 (GBP 300), for the essay <i>Isometric embeddings of surfaces for scl</i> |
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RESEARCH VISITS

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| 3/2026 | Kyōto University (Japan) • One-week visit • invited by RYŌKICHI TANAKA |
| 11/2025 | INI, Cambridge (U.K.) • Three-week visit
Programme: <i>Operators, graphs, groups</i> (participant) |

- 6/2025 | **RIMS, Kyōto University** (Japan) • Two-week small-group workshop
Programme: *Spectrum problems for scl via S-machines* (co-organiser)
- 6/2025 | **KIAS, Seoul** (South Korea) • One-week visit • invited by SANG-HYUN KIM
- 4/2025 | **University of Hawai'i at Mānoa** (U.S.) • Ten-day visit • invited by YASH LODHA
- 4/2022 | **IHP, Paris** (France) • One two-week visit and one three-week visit
and 6/2022 | Programme: *Groups acting on fractals* (participant)
- from 6/2019 | **IMPA, Rio de Janeiro** (Brazil) • Two-month M.Sc. internship
to 8/2019 | Topic: *Topology of complex affine varieties* • Supervisor: HOSSEIN MOVASATI
- from 5/2018 | **KIT, Karlsruhe** (Germany) • Six-week B.Sc. internship
to 6/2018 | Topic: *Geometry of Coxeter groups* • Supervisor: CATERINA CAMPAGNOLO

HIGHER EDUCATION TEACHING EXPERIENCE

- from 1/2023 | **Geometry** • Trinity College & Peterhouse College, Cambridge
to 5/2024 | Part IB supervisions • 34 contact hours • in English
- from 1/2024 | **Geometric Group Theory** • University of Cambridge
to 3/2024 | Part III examples classes • 8 contact hours • in English
- from 10/2022 | **Analysis & Topology** • Trinity College & Peterhouse College, Cambridge
to 1/2024 | Part IB supervisions • 46 contact hours • in English
- from 12/2022 | **Admissions interviews** • Christ's College, Cambridge
to 12/2023 | Co-interviewer for undergraduate admissions in mathematics • in English
- from 1/2023 | **Group Cohomology** • University of Cambridge
to 5/2023 | Part III drop-in sessions • 4 contact hours • in English
- from 1/2022 | **Riemann Surfaces** • Trinity College & Peterhouse College, Cambridge
to 5/2022 | Part II supervisions • 31 contact hours • in English
- from 10/2021 | **Linear Algebra** • Trinity College, Cambridge
to 1/2022 | Part IB supervisions • 16 contact hours • in English
- from 10/2021 | **Mapping Class Groups** • University of Cambridge
to 1/2022 | Part III drop-in sessions • 4 contact hours • in English
- from 9/2018 | **Colles de Mathématiques** • Lycée Aux Lazaristes, Lyon
to 5/2019 | Oral examinations in mathematics for first-year students in a preparatory class • 44 contact hours • in French

VOLUNTEER TEACHING

- 3/2025 | **JSPS science dialogue** • Koshi High School, Fukui
Lecture in English and Japanese for a high school science club
- from 10/2023 | **High school project supervision** • Rusanivskyi Lyceum, Kyiv (online)
to 4/2024 | Student: OLHA MYKHAILYK • Topic: *Inversions and Möbius transformations*
Project received the second prize at the city stage in the Junior Academy of Sciences of Ukraine research competition

from 10/2019 to 3/2020	Tutoring • Coleridge Community College & St Faith's School, Cambridge Volunteer tutoring in mathematics in secondary schools, under the <i>Stimulus</i> programme
from 9/2017 to 6/2020	French as a foreign language • ItinérENS, Lyon French classes for refugees • more than 100 contact hours

SERVICE

Refereeing and quick opinions for *Algebr. Geom. Topol.*, *Enseign. Math.*, *KIAS Springer Ser. Math.*, *Trans. Amer. Math. Soc.*

Reviewing for *zbMATH Open*

from 9/2025 to 9/2027	Analytic & geometric group theory seminar • IM PAN, Warszawa Co-organiser
9/2025	World of GroupCraft V • Online Co-organiser
6/2025	RIMS Workshop • RIMS, Kyōto University Topic: <i>Spectrum problems for scl via S-machines</i> • Co-organiser
from 2/2025 to 4/2025	Reading seminar • Kyōto University Topic: L^2 -Betti numbers • Organiser
from 9/2023 to 8/2024	Junior geometry seminar • University of Cambridge Co-organiser
from 8/2023 to 5/2024	MCR committee • Trinity College, Cambridge MCR (Middle Combination Room) committee member in charge of: the MCR's website, mailing list, elections, lunchtime seminars, etc.
from 10/2022 to 12/2022	Reading seminar • University of Cambridge Topic: <i>Generalisations of hyperbolic groups</i> • Organiser

INVITED TALKS (SELECTION)

10/2026	Università di Pisa / Scuola Normale Superiore LiT IV Landing in Topology
6/2026	ICMAT, Madrid Advanced school: Aspherical 2-complexes and their fundamental groups: past, present, and future
6/2026	Université Bourgogne Europe Séminaire géométrie, algèbre, dynamique et topologie
6/2026	Université de Lorraine Séminaire de géométrie différentielle
11/2025	Newcastle University Algebra & geometry seminar
11/2025	University of Cambridge Geometric group theory seminar
7/2025	University of Tōkyō Tuesday seminar on topology

7/2025	Tōhoku University Geometry seminar
6/2025	KIAS, Seoul Topology seminar
4/2025	University of Hawai'i at Mānoa Mathematics department colloquium
11/2024	Kyōto University First MMS workshop for young researchers
3/2024	University of Oxford Geometric group theory seminar
4/2023	KU Leuven Methusalem junior seminar
3/2023	ETH Zürich Geometry graduate colloquium

LANGUAGES

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| <ul style="list-style-type: none"> • French • Native language • English • Fluent (C2)
Passed the CAE (Cambridge) • Japanese • Advanced (B2)
Passed the JLPT N2 | <ul style="list-style-type: none"> • German • Advanced (B2) • Portuguese • Intermediate (B1) • Polish • Intermediate (A2/B1) • Russian • Intermediate (A2/B1) |
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September 2, 2026